Prove trigonometric identities

1 Prove the following identities:

a $\dfrac{\sin x}{\cos x \tan x} = 1$

b $\dfrac{\cos x \tan x}{\sin x} = 1$

c $\sin A \cos A \tan A = \sin^2(A)$

d $\dfrac{\sin x \cos(90^\circ - x)}{\cos x \sin(90^\circ - x)} = \tan^2(x)$

e $\dfrac{\sin^2(x)}{\cos^2(x)} = \tan^2(x)$

2 Prove the following identities:

a $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$

b $\cos^4(x) - \sin^4(x) = 2\cos^2(x) - 1$

c $2\cos^2(\theta) - 3 = -1 - 2\sin^2(\theta)$

d $5\cos^2(\theta) - 3 = 2 - 5\sin^2(\theta)$

e $(1 + \tan^2(\alpha))\cos^2(\alpha) = 1$

f $\tan^2(y) - \sin^2(y) = \tan^2(y)\sin^2(y)$

3 Prove the following identity:

$$\sin^2(a) - \sin^2(b) + \cos^2(a)\sin^2(b) - \sin^2(a)\cos^2(b) = 0$$

4 Prove the following identities:

a $\dfrac{\sin x \cos x}{\tan x} = \cos^2(x)$

b $\dfrac{1 - \cos^2(\theta)}{\sin\theta} = \sin\theta$

c $\dfrac{1 - \sin^2(x)}{\cos x} = \cos x$

d $\dfrac{1 - \sin^2(x)}{\cos x} = \cos x$

e $\dfrac{\sin\theta}{1 - \cos\theta} = \dfrac{1 + \cos\theta}{\sin\theta}$

f $\dfrac{\tan^2(x) - 1}{\tan^2(x) + 1} = 1 - 2\cos^2(x)$

g $\dfrac{(1 + \sin\theta)^2 + \cos^2(\theta)}{1 + \sin\theta} = 2$

h $\dfrac{\sin^2(x) + \sin x \cos x}{\cos^2(x) + \sin x \cos x} = \tan x$

5 Prove the following identities:

a $\dfrac{\sec\theta}{\tan\theta} = \csc\theta$

b $\dfrac{1 - \cot x}{1 + \cot x} = \dfrac{\tan x - 1}{\tan x + 1}$

c $\dfrac{\cos\theta}{1 - \sin\theta} - \tan\theta = \sec\theta$

d $\dfrac{\cos\alpha}{1 + \sin\alpha} = \sec\alpha(1 - \sin\alpha)$

e $\tan\theta \sin\theta + \cos\theta = \sec\theta$

f $\cot\theta\left(\sec^2(\theta) - 1\right) = \tan\theta$

g $\dfrac{4 + \tan^2(x) - \sec^2(x)}{\csc^2(x)} = 3\sin^2(x)$

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6 Prove the following identity:

$$\sin\theta\,(1+\tan\theta)+\cos\theta\,(1+\cot\theta)=\frac{\sin\theta+\cos\theta}{\sin\theta\cos\theta}$$

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